Logbook

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Week One

- Read the Grand Challenges and discussed them with my teammates.

- Began to choose our idea, through research and meetings.

Week Two

- Discussed the chosen idea with my teammates.

- Formed the base of our proposal for submission, achieving a leap in general progress in comparison to other teams.

- Began work on the first prior solution “Sundrop Farms.”

Week Three

- Finished the prior solution “Sundrop Farms”, **part of which is shown in figure 1.**

The Sundrop Farms

Established in 2016, the Sundrop Farms facility, **shown in figure 1** grows 17,000 tons of tomatoes a year 15% of Australia’s total crop in ostensibly arid land. It uses fresh irrigation water produced by an on-site desalination plant, powered by sunlight from a network of 23,000 mirrors directed at a 123-metre-high solar tower. The plant, which employs around 300 people, doesn’t need soil; the 180,000 tomato plants are grown hydroponically in stacks, using the 1 million liters of water processed a day through desalination. The farms were built as a way to revolutionize the agricultural system in Australia while utilizing the saline water sources widely available in the country for agriculture and aquaponics production in addition to electricity generation.



Figure 1

Figure 1

- Began writing the problem to be solved.

- Researched the prototype heating, cooling, air humidity, and soil moisture mechanisms.

Week Four

- Went to the first field trip to “Al Tamyuz” Research Center which was used in deciding the shape of the prototype & plant types.

- Finished the problem to be solved, **part of which is shown in figure 2.**

Problem To Be Solved

Solving the decrease in agricultural production due to inefficient cultivation system in Egypt.

According to FAO, Egypt has the lowest hectares per capita value of 0.05 hectares per head. Compared to the global average 0.20 hectares / capita and the recommended global average of 0.50 hectares per capita, Egypt falls behind with a significant amount in terms of food security. However, the value of 0.05 arable hectares per person dates back to the early 2000’s as does the FAO article. Instead, recent reports as early as 2023 from ourworldindata.org report a value as low as 0.03 hectares per person **as shown in figure 1.** The ever-decreasing value of arable hectares of land available per person stems from and is fueled by the condensed population growth on only 6% of Egypt’s land mass, which happens to be the very same arable land that is decreasing at the same time the number of people increases.

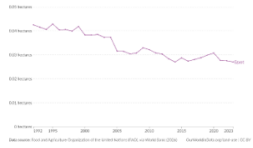


Figure 1, Arrable land use per capita in Egypt 1992-2023

Figure 2

- Wrote the selection of prototype, **part of which is shown in figure 3.**

Selection of Prototype

The selected prototype is an agricultural closed-loop system which provides the required parameters, which are temperature, humidity and soil moisture with the addition of daylight LEDs for maximum growth for the selected crops which are cauliflower and tamarillo within its range. This semester’s challenge imposes that all non-electronic materials are recycled materials. Therefore, all the materials used in the prototype will be extracted from said sources. These chosen sources for the prototype have been selected to be recycled wood, PVC greenhouse plastic and the electrical circuit. The most notable of said components is the ESP32 paired with the Relay-Actuator systems. This circuit serves as the main driver of the entire prototype as it controls when to switch different systems on/off and keeps track of parameters by feeding them through a database. Another main component would be the heat fans, PTC heater plates and the temperature sensors. The fans help distribute heat

Figure 3

Week Five

- Created the 3D design of the prototype. **Which is shown in figure 4.**

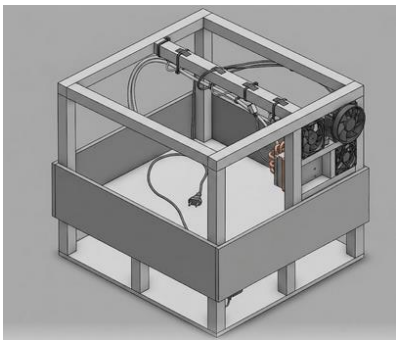


Figure 4

- Chapter I & II were finish as according to the announced capstone Calander.

- Researched the power source of the prototype and began constructing the schematic, exceeding the set scope of our team in the prototyping phase.

Week Six

- Pinpointed and researched all of the prototype's materials encompassing sensors, actuators, and mechanisms.

- Discussed the prototype's design with my teammate which he used to construct the wooden frame of the prototype.

- Constructed a make-shift water tank level system schematic, **as shown in figure 5.** Also reused my previous heating system. With the addition of new cooling system inspired by desert air conditioners.

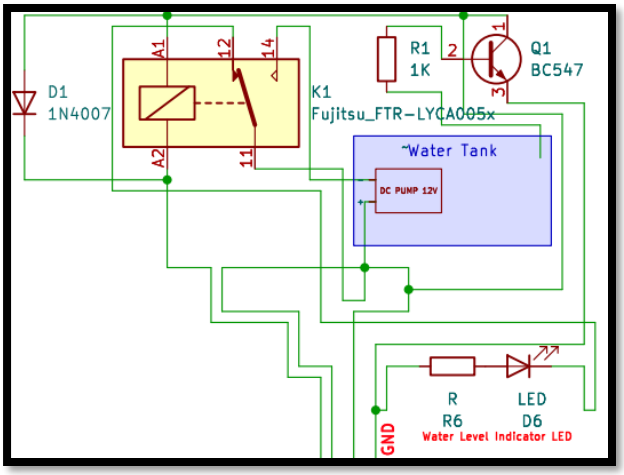


Figure 5

Week Seven

EID Al Fitr WEEK

This week had minimal prototype design advancements and improvements to the general schematic and prototype concept, and mid-term preparation towards the end of the week rather than static progress.

Week Eight

MIDTERM WEEK

This week had general team communication and meetings to determine our next steps in the project, especially the prototype, in addition to schematic concepts improvement.

Week Nine

- Wrote the recommendation about the real-life application **as shown in figure 6.**

The real-life application of this prototype is recommended to be constructed in STEM High School for boys - 6th of October, **which is shown in figure 1**, as it is suitable due to its environmental conditions for our project. It will utilize an integrated cooling system similar to the prototype with the exception of the water spray mechanism being integrated behind the fans themselves. The heating system will utilize the same concept with more spacious heating coils. The rest of the system will remain largely as is. In terms of dimensions, it will follow a scale of 1:10, the height would be 7 meters, and two square bases of segments 5 meters each.



Figure 1

Figure 6

- Attended a capstone meeting with the capstone leader.
- Purchased the selected materials with my teammate.
- Began the construction of the prototype, which is the early phase where the circuit foundations were connected.
- Explained the problem to be solved to my teammate so that he writes a more thorough poster introduction.

Week Ten

- Outperformed in the prototype mid and late phases as it was entirely assembled and integrated into the frame **as shown in figure 7**, the final phase.

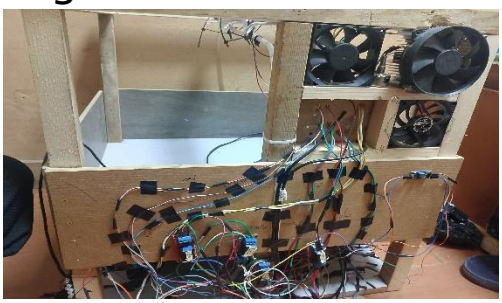


Figure 7

- Completed the full closed-loop code for the prototype, part of which **is shown in figure 8.**

```
#include <WIFI.h>
#include "time.h"
#include <OneWire.h>
#include <DallasTemperature.h>
#include <DHT.h>
#include <ESPAsyncWebServer.h>
// ================== CONFIGURATION ==================
const char* ssid     = "veryhotspot";
const char* password = "catowl101";

const char* ntpServer = "pool.ntp.org";
const char* TZ_INFO = "EET-2EEST,M4.5.5/0,M10.5.4/24";

// ================== PIN DEFINITIONS ==================
#define H_1       4
#define PU_1      17
#define LED_1     21
#define HU_SYS    18
#define VNT_1     16   // active LOW

#define TMP_2_PIN  5
#define TMP_1_PIN  22
```

Figure 8

- Provided information for my teammate putting together the materials table.
- Constructed the database and website systems and the data logging mechanism via local hosting, CSV and Data queries.
- Conducted the test plan and collected the raw results.
- Wrote the poster results and designed the graphs according to the collected data.
- Wrote the methods and test plan sections in the poster.

Week Eleven

- Finalized the schematic design by adding all the later-added modifications, **the full diagram is shown in figure 9.**

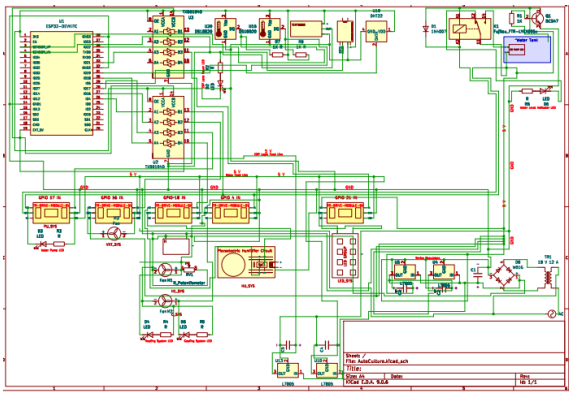


Figure 9

- Attended a capstone meeting with the capstone leader.

- The portfolio, full-size poster and A4 poster were submitted this week.

- Edited the raw test plan videos, (timelapse).